一. 填空

1. B

2. $(165)_{10}=(10100101)_2=(010010011000)_{BCD码}$

3. $\overline{A}(\overline{B}+\overline{\overline{C}+D})+\overline{C}$

4. $[+1101]_{补}=(01101)$ 正数补码.反码.原码相同.

5. $26+20-1=45=32+8+4+1$ 1011011

6. 3V, 5V

7. 4片 (书上扩展用35片选择器. 但最后一片不是必须的, 可以用门电路实现)

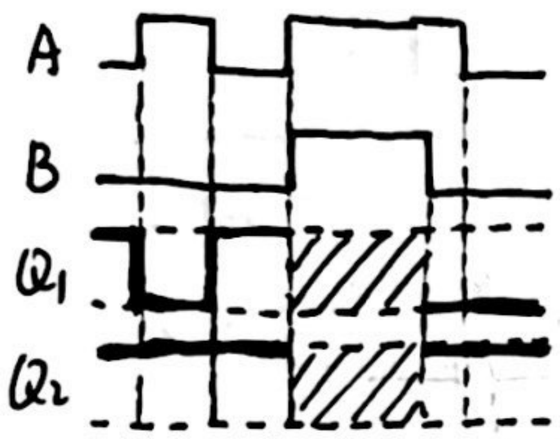
8. 8进制加法计数器

9. D.

10. 11011001 $25=8\times3+1$

11. A

12.



A	B	Q_1	Q_2
1	0	0	1
0	0	1	1
1	1	1/0	0/1
1	0	0	1

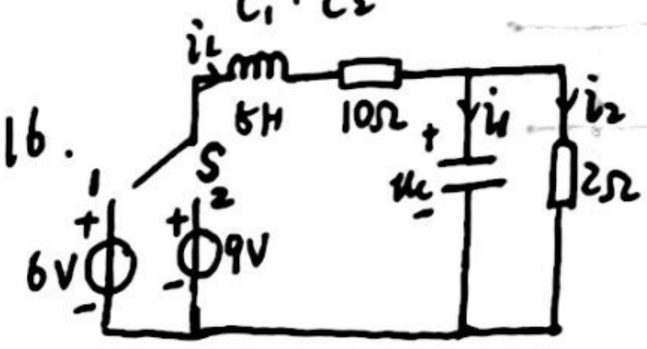
其实就是扯直的RS触发器. 因为用3禁用的00输入会导致结果混乱. (不necessarily用虚线表示).

13. $2^{10}\times4=4096$

14. RAM

15. $C=\frac{1}{\frac{1}{C_1}+\frac{1}{C_2}}=50\mu F$. $R=2\sqrt{\frac{L}{C}}=20\Omega$

16.



$i_L=i_1+i_2=i_C+\frac{u_C}{2}=C\frac{du_C}{dt}+\frac{1}{2}u_C=0.2\frac{du_C}{dt}+0.5$

$u_L+i_L\cdot10\Omega+u_C=9$

$\Rightarrow L\frac{di_L}{dt}+10i_L+u_C=9 \Rightarrow \frac{d^2u_C}{dt^2}+4.5\frac{du_C}{dt}+6u_C-9=0$.

$\Delta=4.5^2-4\times1\times6u_C$

$u_C(0^+)=u_C(0^-)=6V$, $\Delta=4.5^2-4\times1\times6<0$. ∴欠阻尼.

二. 简答题 1. $V_{BE}=0.7V$ 三极管导通.

$I_B=\frac{V_1-V_{BE}}{2k\Omega}=2.15mA$.

假设放大.

$I_C=\beta I_B=107.5mA$.

$V_{CE}=15-I_C\cdot5k\Omega<0$ ∴假设不成立. 在处于饱和.

∴ $V_{CE}=V_{CES}=0.3V$

$I_C=\frac{15V-0.3V}{5k\Omega}=2.94mA$.

2. $Z=\Sigma m(0,1,5,10)+\Sigma d(2,3,4,6,11,12,13)$

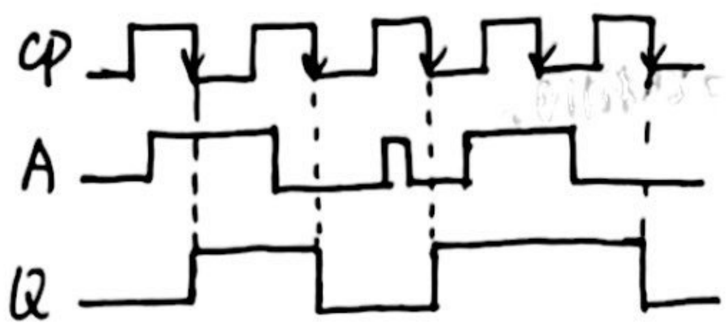
$Z=\overline{A}\,\overline{C}+\overline{B}C$

3. $Z_1=AB+\overline{B}A\overline{C}$

$Z_2=\overline{A}\,\overline{B\overline{C}D}$

4. $Q^{n+1} = J\overline{Q^n} + \bar{K}Q^n = \overline{Q^n}\cdot\overline{Q^n} + \bar{\bar{A}}Q^n = \overline{Q^n} + AQ^n = \overline{Q^n} + A$

CP	A	Q^n	Q^{n+1}
0	×	/	0
1↓	1	0	1
2↓	0	1	0
3↓	0	0	1
4↓	1	1	1
5↓	0	1	0



5. 真值表略. 卡诺图

A \ BC	00	01	11	10
0	0	0	1	0
1	1	1	1	0

$Z = A\bar{B} + BC$

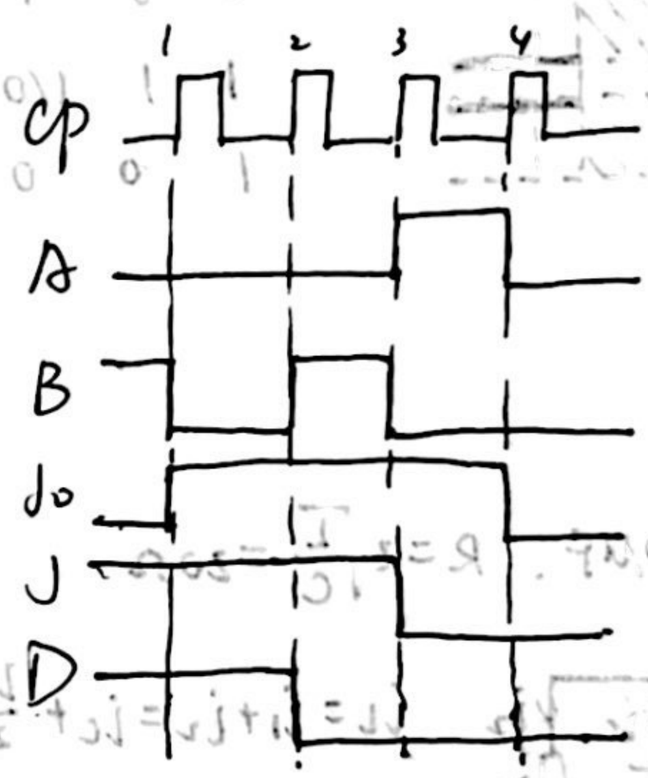
三. (1) 真值表略.

(2) $Z = \Sigma m(4,8,12) + \Sigma d(0,15)$

(3) $Z = \overline{A_1}\,\overline{A_0}$

四. (1)

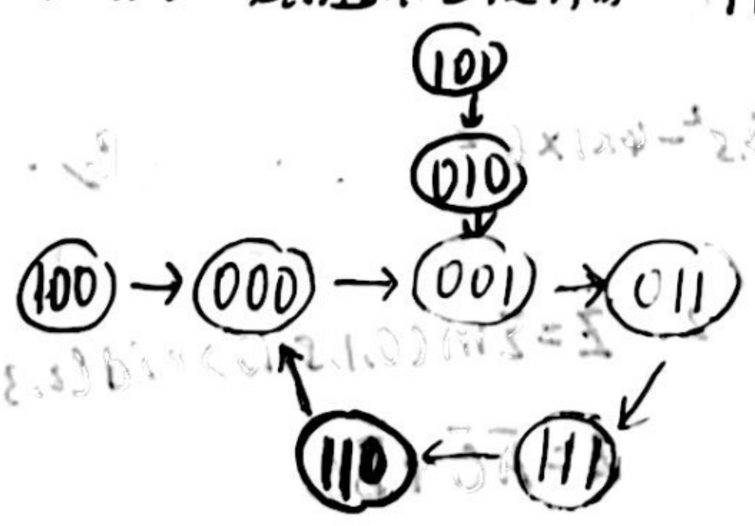
CP	A B	J_0	D J	SR_3
0	0 1	0	1 1	0 0 0 0
1↑	0 0	1	1 1	1 0 0 0
2↑	0 1	1	0 1	1 1 0 0
3↑	1 0	1	0 0	0 1 1 0
4↑	0 0	0	0 0	0 0 1 1



五. (1). $Q_2^{n+1} = Q_1^n Q_0^n$, $Q_1^{n+1} = Q_0^n$, $Q_0^{n+1} = \overline{Q_2^n}$

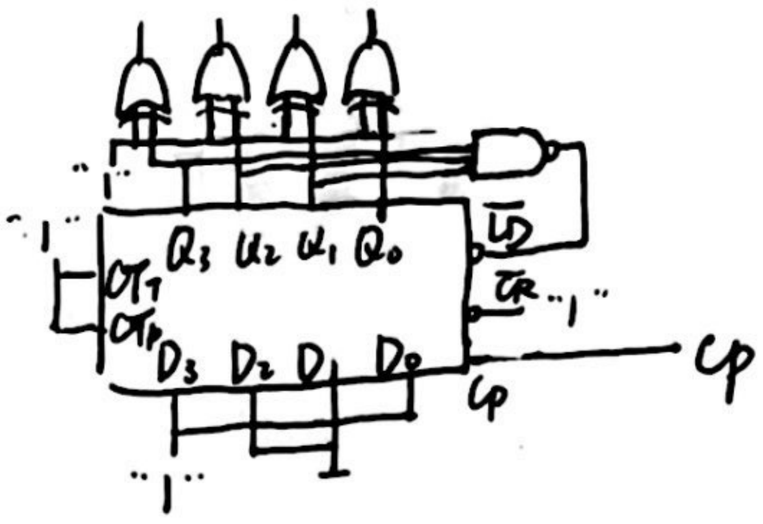
(2) 状态真值表略, 从000画起即可

判断自启动: 真值表里没有的几种状态: 100→000, 101⇒010→001

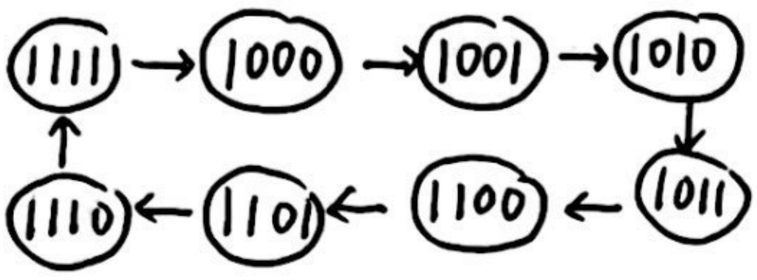


(3) $Z = \overline{Q_1}$, $f_{CP} = 1\,Hz$ ($Z = Q_1$, $Z = Q_0$ 应该也对)

六、 1. 复位后 1001 → 1010 → … → 1110



2. 8进制加法计数器.



3.

